E-Commerce Business Intelligence Suite: Project Report

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Tools Used: SQL, Excel, Python, Power BI

1. Project Objective

The primary objective of the "E-Commerce Business Intelligence Suite" project is to transform raw operational e-commerce data into actionable insights. This project aims to establish a robust business intelligence framework that enables data-driven decision-making, specifically focusing on optimizing sales performance, understanding customer behavior, and identifying key revenue drivers.
This solution empowers stakeholders to make better business decisions and improve performance by utilizing KPI-driven reporting and real-time filtering to track key metrics and identify emerging trends.

2. Dataset Overview

The analysis utilized a comprehensive e-commerce dataset comprising five primary tables, interconnected through unique identifiers.

Table Name	Key Columns Used	Description
Customers	customer_id, customer_state	Information on unique customers and their location.
Orders	order_id, customer_id, order_status, order_purchase_timestamp, order_delivered_customer_date	Details on each order, its status, and time stamps.
Order Items	order_id, order_item_id, product_id, price	Line-item information for each order, including unit price.
Payments	order_id, payment_type, payment_value	Details regarding payment methods and values for orders.
Products	product_id, product_category_name	Information about products, primarily their categories.

3. Workflow / System Design

The project followed a structured five-phase workflow, ensuring data quality, analysis readiness, and effective visualization.

- **Phase 1: Data Extraction (SQL & Python)**
  - Data was extracted from the source relational database using SQL queries.
  - Python was used for initial data profiling and handling large volumes.
- **Phase 2: Data Cleaning and Transformation (Excel & Python)**
  - Missing values, duplicates, and inconsistent data types were addressed.
  - Calculated fields (e.g., AOV, Revenue) and date dimension columns were created.
- **Phase 3: Data Modeling (Excel & Power BI)**
  - A star schema data model was established, connecting fact tables (Orders, Order Items, Payments) to dimension tables (Customers, Products).
- **Phase 4: KPI Development (Power BI)**
  - Key Performance Indicators (KPIs) were defined and calculated using Data Analysis Expressions (DAX) in Power BI.
- **Phase 5: Report and Dashboard Visualization (Power BI)**
  - Interactive dashboards were developed to present findings and allow stakeholders to explore data dynamically.

4. KPIs Tracked

The following Key Performance Indicators were tracked to measure business performance, with filters applied for **Delivered Orders** only to ensure accuracy in sales analysis.

Key KPIs Formula:

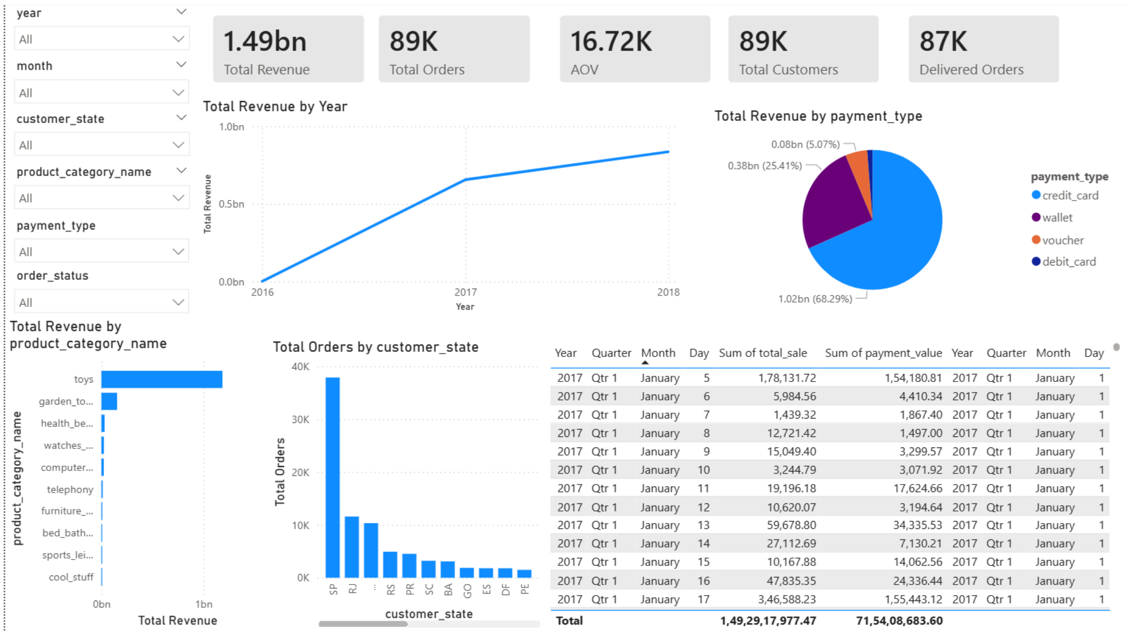
- **Total Revenue = SUM(total_sale)**
- **Total Orders = Distinct count(order_id)**
- **Delivered Orders = Distinct count(order_id) where status = delivered**
- **AOV = Total Revenue / Total Orders**

KPI	Description	Filters Used
Total Revenue	Sum(total_sale) for delivered orders	order_status = 'delivered'
Total Orders	Count of unique delivered orders.	order_status = 'delivered'
Total Customers	Count of unique customers.	N/A

KPI	Description	Filters Used
Delivered Orders	Total count of orders with 'delivered' status.	order_status = 'delivered'
Average Order Value (AOV)	Total Revenue / Total Orders.	order_status = 'delivered'

5. Dashboard Insights & Findings

The Power BI dashboard provided deep, filterable insights into sales performance and customer trends.



Revenue Trend Insight

- The monthly revenue trend revealed a seasonal pattern, with peak sales generally occurring in the second half of the year.
- A significant drop in revenue was observed in **September 2017 and October 2017** warranting further investigation into marketing spend or product availability during those periods.

Category Insight

- **Top category (Toys)** contributes the highest revenue, making it a priority segment for campaigns and inventory planning.
- Categories like **Toys** and **Garden Tools** contributed significantly to total revenue, making them ideal targets for inventory scaling and promotional campaigns.
- The majority of orders originated from two key states: São Paulo (SP) and Rio de Janeiro (RJ).

- Sales volume was notably low in Northern and Central-West states, pointing to underdeveloped market penetration in those regions.

Payment Insight

- Credit Cards were the dominant payment method, accounting for approximately 75% of total revenue.
- Payment methods like Boleto (cash voucher) and debit card, while secondary, still represented a measurable portion of the customer base, suggesting a need to ensure smooth processing for these methods.

6. Business Recommendations

Based on the data-driven insights, the following actionable recommendations are proposed to optimize business performance:

1. **Optimize inventory and marketing for peak demand periods** to prevent missed sales opportunities.
2. **Strengthen top-performing product categories** through bundling, pricing tests, and promotion strategies.
3. **Expand regional market reach** using state-wise campaigns and delivery optimization in low-performing regions.
4. **Improve payment conversion** by offering discounts/EMI for credit cards and incentives for alternative methods.
5. **Investigate revenue decline periods** using deeper analysis (delivery delays, stock-outs, pricing changes, or reduced demand).

7. Conclusion

The E-Commerce Business Intelligence Suite successfully delivered a comprehensive framework for converting raw transactional data into strategic insights. The report identifies clear opportunities for optimizing marketing spend, improving inventory management, and expanding geographic market share. The findings provide a solid foundation for the business to execute targeted strategies and drive sustained revenue growth.

8. Tools & Skills Demonstrated

- **SQL:** Database design, joins, KPI queries, trend analysis
- **Excel:** Data cleaning, pivot tables, KPI dashboard
- **Python:** EDA, data merging, feature engineering, automation exports
- **Power BI:** DAX measures, slicers, interactive dashboard design

9. Attachments / Deliverables List

- Final Power BI Dashboard File: E com bi report.pbix

- Data Transformation Python Script: eda_automation.py
- SQL Data Extraction Queries: ecom.sql